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COMPUTATIONAL GAME THEORY PAPER REVIEW

by

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**A Characterization of Sequential Equilibrium through E-Perfect γ -Sequential Equilibrium
with Local Sequential Rationality and Its Computation**

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Date: 05 June 2026

1 INTRODUCTION

1.1 Summary of the Paper

The Problem Being Addressed

This paper studies how to find a Nash equilibrium in complex, step-by-step games. A Nash equilibrium is a situation where no player can do better by changing their own strategy alone.

Originally, these games were solved by turning them into a large table (a matrix form). But this does not work well for big games because the number of possible strategy combinations becomes extremely large very quickly.

To solve this, researchers developed the sequence form approach, which focuses on the actual paths players can follow instead of listing every possible move. This makes the problem smaller, but it also creates a new issue: one path can represent many different ways a player might behave. Because of this, it is still hard to accurately calculate expected outcomes at each decision point.

The authors address these problems by using a new method that works directly with the game's step-by-step structure. Instead of converting the game into a large table or only using paths, their approach studies each decision point locally. This makes the calculations more stable and easier to solve.

1.2 Main contribution of the work

The paper introduces a new way to find Nash Equilibria in Behavioral Strategies for extensive-form games. Instead of solving one very large and difficult problem, the authors break it into many smaller and easier problems at each decision point. They also introduce self-independent

beliefs, where a player's beliefs depend only on the actions of other players and chance, not on their own past actions. The paper proposes a new idea called the semi-sequential equilibrium and a new algorithm that makes the calculations easier and more efficient. The authors prove that this approach can find Nash Equilibrium without converting the game into normal form. Their experiments also show that the method works well for finding both Nash equilibria and subgame perfect equilibria in multi player games.

2 RELATED WORK

The paper is based on many important studies in game theory. It uses the work of Nash (1950, 1951), who introduced the idea of the Nash equilibrium, and Kuhn (1953), who explained behavioral strategies in extensive-form games. The authors compare their method with the traditional method described by Myerson (1991), which changes extensive-form games into normal-form games. However, this approach becomes very difficult for large games because the number of possible strategies grows rapidly. They also discuss the sequence form method by Koller and Megiddo (1992) and von Stengel (1996), which reduces the size of the game but still has difficulty calculating expected payoffs at different decision points.

The paper also builds on earlier research about equilibrium refinements and belief systems. It uses Selten's (1965, 1975) ideas of subgame perfect and perfect equilibrium, which explain how players should make decisions even in situations that are unlikely to happen. The authors also use ideas from Kreps and Wilson (1982) and Fudenberg and Tirole (1991) on sequential and perfect Bayesian equilibrium. They extend the work of Mas-Colell et al. (1995) by providing a simpler way to describe Nash equilibrium without solving one large and complicated optimization problem. Finally, they compare their method with earlier algorithms developed by Wilson (1972) and Herings and Peeters (2001), showing that those earlier methods were mainly designed for games with only two players.

3 LIMITATIONS OF THE PAPER

Although the paper makes an important contribution to game theory, it has some limitations.

First limitation

First, the proposed method only works for games with perfect recall, where players always remember their previous actions and information. It may not work well for games with imperfect recall, where players can forget earlier decisions.

Second limitation

Secondly, even though the method avoids the large complexity of the normal-form approach, it still requires solving complex polynomial equations. For very large game trees, solving these equations can still be computationally expensive and difficult.

Third limitation

Finally, the proposed path-following algorithm uses a logarithmic-barrier method to solve the problem. However, this method can become unstable and depends a lot on the starting point and step size, especially when strategy probabilities are very close to 0 or 1.

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